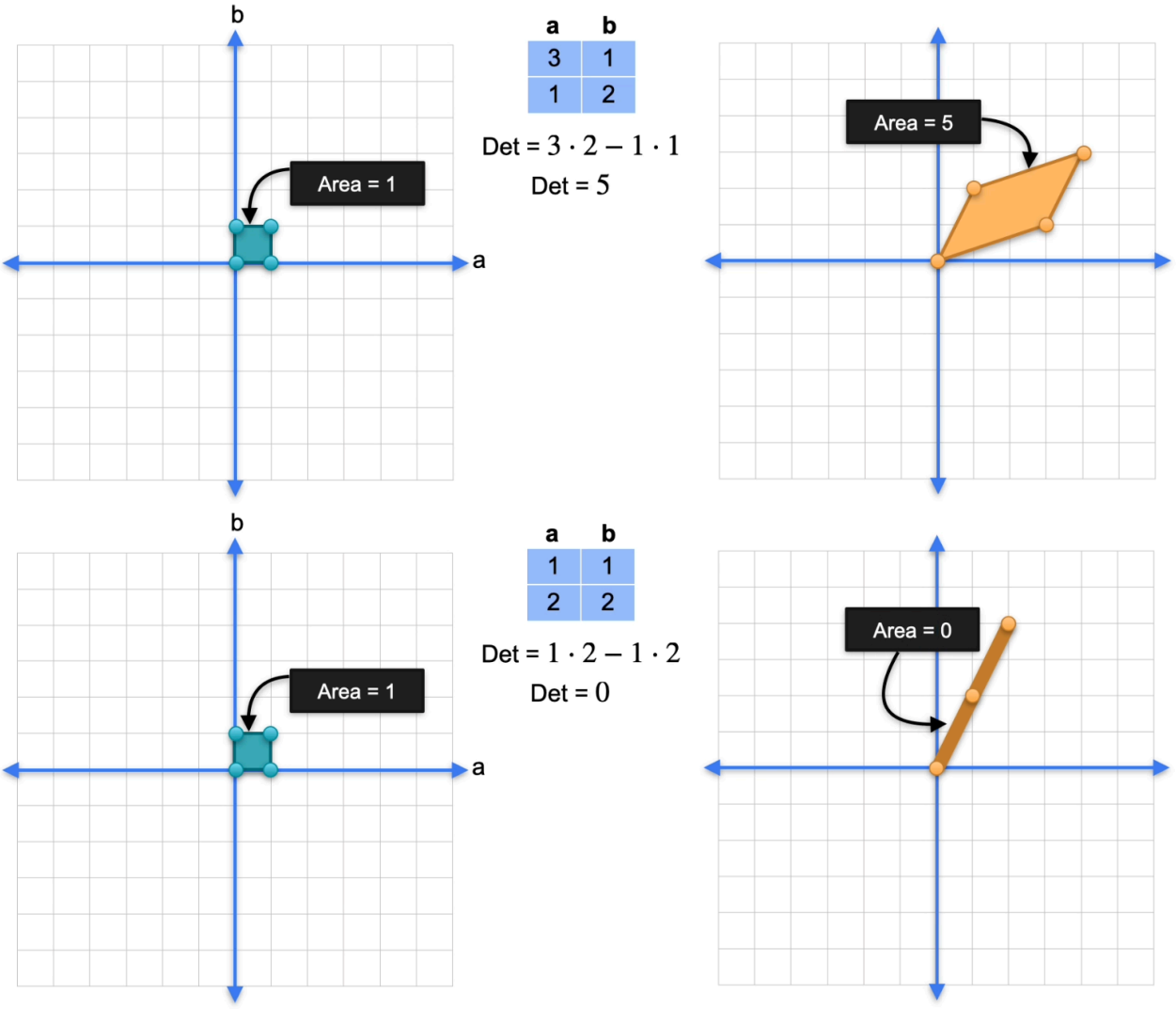
Determinant as an area

Recall: The determinant is a way to tell whether the matrix is singular or non-singular.

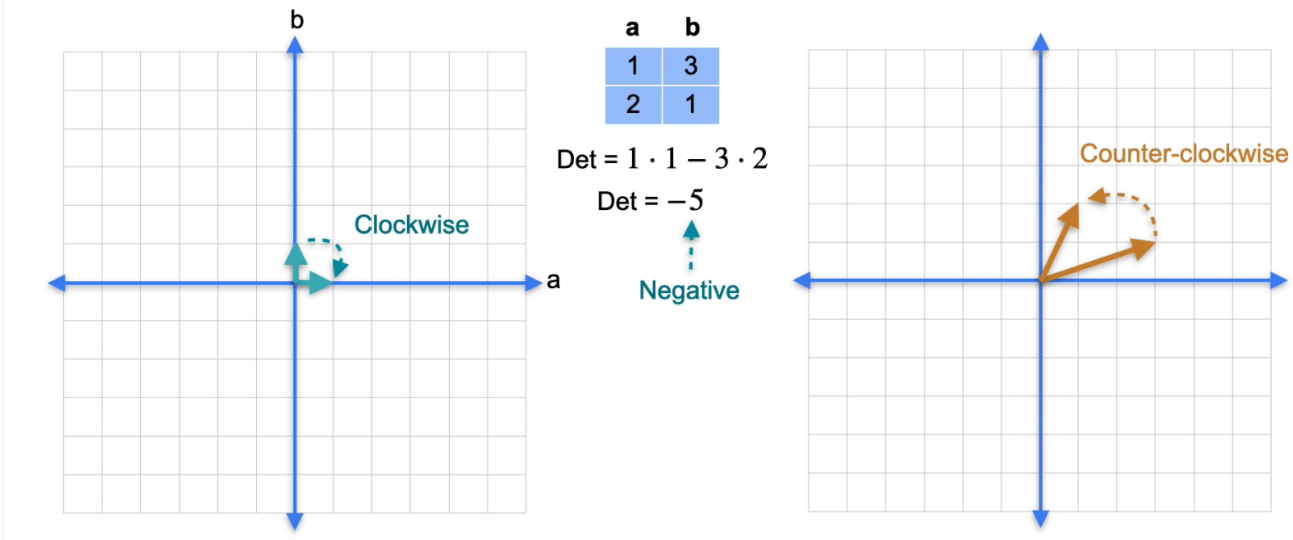
The determinant as an area, when the area or volume is zero, then the matrix is singular.

The determinant of the matrix is the area of the image of the fundamental basis formed by the unit square on the left.



If the matrix is singular, the area = 0.

Negative Determinant



Determinant of a product

3	1	1	1	=	1	4
1	2	-2	1		-3	3

det = 5 det = 3 det = 15

$3 \cdot 2 - 1 \cdot 1$ $1 \cdot 1 - 1 \cdot (-2)$ $1 \cdot 3 - 4 \cdot (-3)$

The product of a singular and a non-singular matrix (in any order) is: singular
if A is non-singular and B is singular, then det(AB) = det(A) x det(B) = 0, since det(B) = 0, so The AB is **singular**

Determinant of an inverse

3	1	-1	=	0.4	-0.2	5	2	-1	=	0.25	-0.25	1	2	=	?	?
1	2			-0.2	0.6	1	2			-0.125	0.625	1	2		?	?

det = 5 det = 0.2 det = 8 det = 0.125 det = 0 det = ???

$5^{-1} = 0.2$ $8^{-1} = 0.125$ $0^{-1} = ???$

If the matrix is non-singular (inversible), then

$$det(A^{-1}) = \frac{1}{det(A)}$$

because

$$det(AB) = det(A)det(B)$$

$$det(AA^{-1}) = det(A)det(A^{-1})$$

$$det(I) = det(A)det(A^{-1}) = 1$$